

Ogan K. Kumova, PhD

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TEACHING PHILOSOPHY

I teach science as a process of discovery rather than a collection of facts to memorize. In my classroom, I want students to understand not only what is happening, but why — and how they can work through it themselves. I rely on guided discussion, Socratic questioning, collaborative activities, and real-world case studies. I also deliberately avoid anchoring my courses to a single textbook — drawing instead from primary literature, case studies, and open educational resources. As zero-cost course initiatives like Z-courses and OER adoption become more widespread, I believe removing textbook barriers is one of the most practical ways faculty can support equitable access to education.

My teaching has spanned a wide range of learners — community college students returning to school after years away, undergraduates entering health professions, graduate and medical students, postdoctoral fellows, and neonatology fellows. At Rowan College at Burlington County, I run the Health and Human Microbiology course for health-profession students and also teach Biology I, Biology II, and Basic Microbiology. At Drexel University College of Medicine and Uniformed Services University, I lecture in microbiology, immunology, molecular pathogenesis, and vaccine development.

Good teaching, to me, is not about simplifying material until it loses its depth. It's about presenting it with clarity, structure, and relevance — and making sure students feel supported enough to engage actively with challenging ideas. My goal is for students to leave not just knowing more biology, but better equipped to question, analyze, and think for themselves.

TEACHING EXPERIENCE

Senior Adjunct Faculty

August 2020 – Present

Rowan University / Rowan College at Burlington County — STEM Division

- Creator and lead instructor for BIO 112: Health and Human Microbiology — designed from the ground up for students in health-profession tracks; curriculum integrates microbial pathogenesis, clinical case studies, and evidence-based reasoning.
- Teaches BIO 155: Basic Microbiology; BIO 120: Biology I; BIO 122: Biology II.
- Delivers all courses across in-person, hybrid, and fully asynchronous online formats, maintaining consistent academic rigor and student engagement across modalities.
- Deliberately uses open educational resources and zero-cost course materials in place of commercial textbooks to eliminate financial barriers to student success.
- Advises students on transfer planning, health-profession pathways, and academic skill development.
- Recognized for teaching excellence: Excellence in Teaching Award, Rowan College at Burlington County, 2022 & 2023.

Visiting Lecturer

March 2022 – 2025

Uniformed Services University Medical College

- Delivered invited graduate and medical school lectures in Molecular Pathogenesis for military physicians and scientists.

Lecturer & Graduate Teaching Assistant

March 2015 – Present

Drexel University College of Medicine — Graduate School of Biomedical Sciences

- Lectures for PhD and MD students in: Vaccine Development (MIIM-524, 2019–Present); Medical Microbiology; Immunology I (MIIM-607, 2016–Present) — Antiviral Immunity, CD8+ T Cells, NK Cells, Apoptosis; Introduction to Infectious Diseases (Spring 2016); Clinical Correlations in Infectious Diseases (Fall 2015, 2016); Clinical Correlation in Immunology (Spring 2017).
- Teaching Assistant, MIIM-512 & MIIM-521: Molecular Pathogenesis I & II.

- CRLA Level II Certified Tutor; provided structured, evidence-based academic mentorship to graduate and professional students.
- Mentored undergraduate, graduate, medical, postdoctoral, and neonatology fellows in research design, scientific writing, and career development.

Adjunct Faculty

January 2012 – June 2013

Drexel University — School of Arts and Sciences & Goodwin College of Professional Studies

- Chemistry II (lecture); Physical Chemistry Lab; Organic Chemistry Lab; Analytical Chemistry/Instrumental Analysis; Chemistry for Engineers; Chemistry for Health Science.

Adjunct Faculty

December 2010 – 2015

Philadelphia University — School of Science and Health

- Chemistry I & II (lecture and laboratory).
- Designed and developed curriculum for chemistry laboratories and the undergraduate course Critical Evaluation of Primary Scientific Literature — emphasizing active reading, experimental design, and peer review.
- Consulted on the physical remodeling and redesign of laboratory facilities to support modern, hands-on pedagogy.

Teaching Assistant & Instructor

August 2007 – September 2010

University of the Sciences — Departments of Chemistry & Biochemistry and Biological Sciences

- Lead Instructor: General Chemistry Laboratories CH 102 & CH 104; Critical Evaluation of Primary Scientific Literature.
- Teaching Assistant: Principles of Chemistry Lab CH 113 & CH 114; Biochemistry Lab CH 346; Chemical Sciences Orientation CH 110.
- Guest Lecturer: Discussions in Chemistry (upper-division students). Forensic Anthropology laboratory TA (held at the University of Pennsylvania Museum of Archaeology and Anthropology).

Museum Educator, Program Developer & Acting Camp-In Director

May 2003 – March 2020

Franklin Institute Science Museum — Philadelphia, PA

- Delivered interactive, inquiry-based science programming to public audiences of all ages; developed and performed traveling science shows for off-site venues across the Philadelphia region.
- Designed workshop curricula for K–12 school groups and summer camp programs, fostering scientific curiosity and literacy across age groups and learning levels.
- Served as Acting Director of the overnight camp-in program: managed staff, budgets, and logistics for events hosting up to 400 participants twice per week during the school year.
- Contributed to the planning, development, and execution of the Philadelphia Science Festival.

EDUCATION

Doctor of Philosophy, Microbiology & Immunology

August 2015 – July 2020

Drexel University College of Medicine — Philadelphia, PA

GPA: 4.0 · Dissertation: Altered Innate Immune Responses to Influenza in the Neonate · Minor: Health Administration

Master of Science, Infectious Diseases

January 2013 – May 2015

Drexel University College of Medicine — Philadelphia, PA

GPA: 3.97

B.S. in Biochemistry & B.S. in Bioinformatics

August 2005 – May 2009

University of the Sciences in Philadelphia — Philadelphia, PA

Minor in Mathematics

PROFESSIONAL BACKGROUND

Program Officer, Vaccine Safety

February 2023 – Present

National Academies of Sciences, Engineering, and Medicine — Washington, D.C.

- Manages expert committee studies on vaccine safety for federal agencies (CDC, VA, DoD, FDA), overseeing full study lifecycle: expert recruitment, committee coordination, evidence synthesis, and report production.
- Contributing Editor on two published National Academies reports (National Academies Press, 2024 & 2025).
Delivered invited Congressional briefings on vaccine safety (May 2024; October 2025).

ORISE Postdoctoral Research Fellow

October 2020 – February 2023

U.S. Food and Drug Administration — Silver Spring, MD

- Developed *Neisseria gonorrhoeae* vaccine candidates (liposome and OMV platforms) under the NIH-U19 Gonorrhea Vaccine Cooperative Research Center. PI: Margaret C. Bash, MD, MPH.

SELECTED PEER-REVIEWED PUBLICATIONS

15 peer-reviewed articles. * denotes co-first authorship. Selected publications below.

Rao AS, Onufer AP, **Kumova O**, Roberts LTN, Mossabeh R, Pascasio J, Carey AJ. Myeloid TLR2 signaling amplifies immunopathology in influenza-infected murine neonates. *Am J Respir Cell Mol Biol*. 2026;aanag050. PMID: 42089286.

Kumova OK, Galani IE, Rao A, Johnson H, Triantafyllia V, Matt SM, Pascasio J, Gaskill PJ, Andreacos E, Katsikis PD, Carey AJ. Severity of neonatal influenza infection is driven by type I interferon and oxidative stress. *Mucosal Immunol*. 2022;15(6):1309–1320. PMID: 36352099.

Kumova OK, Fike AJ, Thayer JL, Nguyen LT, Mell JC, Pascasio J, Stairiker C, Leon LG, Katsikis PD, Carey AJ. Lung transcriptional unresponsiveness and loss of early influenza virus control in infected neonates is prevented by intranasal *Lactobacillus rhamnosus* GG. *PLoS Pathog*. 2019;15(10):e1008072. PMID: 31603951.

Kumova OK*, Fike AJ*, Carey AJ. Dissecting the defects in the neonatal CD8+ T-cell response. *J Leukoc Biol*. 2019;106(5):1051–1061. PMID: 31260598. (Review)

Matthias KA, Connolly KL, Jerse AE, **Kumova OK**, Macintyre AN, Gray MC, Thomas KS, Criss AK, Zielke RA, Li L, Sikora AE, Ramirez-Bencomo F, Thistlethwaite A, Derrick JP, Lu WE, Bash MC. Evaluation of immunization route in induction of vaccine-mediated anti-gonococcal immune responses in a murine model of ascending infection. *J Infect Dis*. 2025;232(5):e765–e777. PMID: 40971931.

Full publication list: https://pubmed.ncbi.nlm.nih.gov/?term=Kumova%20O&cauthor_id=42089286

National Academies Reports — Contributing Editor

Ancona D, **Kumova OK**, Stratton K, Henney JE, editors. Vaccine Risk Monitoring and Evaluation at the Centers for Disease Control and Prevention. Washington, DC: National Academies Press; 2025. doi:10.17226/29240. PMID: 41289413.

Rosenberg D, **Kumova OK**, Stratton K, Bass AR, editors. Evidence Review of the Adverse Effects of COVID-19 Vaccination and Intramuscular Vaccine Administration. Washington, DC: National Academies Press; 2024. doi:10.17226/27746. PMID: 39312602.

TEACHING CERTIFICATIONS & CREDENTIALS

CRLA (College Reading and Learning Association) Level II Certified Tutor · Blackboard, Canvas, Kaltura — online and hybrid course delivery · Open Educational Resources (OER) & Z-Course Design · Bilingual: Turkish and English · First Aid & CPR